

Euclid's Elements — Book II

Proposition 6

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If a straight line is bisected and then produced in a straight line, the rectangle contained by the whole with the added segment and the added segment, together with the square on the half, equals the square on the line made up of the half and the added segment.

1 Introduction

Proposition II.6 is the external-section companion of Proposition II.5. A straight line AB is bisected at C and then produced beyond B to D . Thus the one-dimensional order used by the formal theorem is

$$A - C - B - D.$$

The proposition asserts, in geometric content language,

$$\text{Sq}(BC) + \text{Rect}(AD, BD) = \text{Sq}(CD).$$

The familiar modern algebraic shadow is obtained by writing the half-length $AC = CB$ as a and the added segment BD as x :

$$a^2 + (2a + x)x = (a + x)^2.$$

Equivalently, if $y = CD$ and $z = CB$, one sees the difference-of-squares pattern

$$(y + z)(y - z) + z^2 = y^2.$$

Both formulas are documentary mnemonics only. The Lean proof does not define a multiplication of segment lengths and does not evaluate a numerical area. Its objects are concrete rectangle and square representatives, and the conclusion is exact scissors equality.

The production source begins with

```
import CGJteamLab.Proposition2_5
```

so Proposition II.5 is a module dependency. It is not, however, a theorem dependency of the proof term. The current II.6 proof directly calls the binary theorem from II.1, Proposition II.3, Proposition II.4, and the rectangle-transport/scissors API. This distinction between the module chain and the actual theorem-call DAG is important in the present reconstruction.

2 Euclid's Statement and Classical Configuration

2.1 The statement

In Heath's wording, the proposition begins:

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.

With AB bisected at C and BD added in a straight line, the concrete statement is

$$\text{Rect}(AD, DB) + \text{Sq}(CB) = \text{Sq}(CD) \quad [?].$$

As throughout the present Book II documentation, the notation $\text{Rect}(-, -)$ and $\text{Sq}(-)$ is documentary shorthand for geometric figures. It is not a definition of a numerical product or numerical square.

2.2 The classical diagram

Euclid constructs the square $CEFD$ on CD , joins the diagonal DE , draws through B a parallel to CE (or DF), then through the resulting intersection H a line parallel to the baseline, and finally through A a parallel to the vertical sides. In the annotated diagram we also mark N as the midpoint of BD , so that the traditional name of the gnomon NOP has a literal set of boundary labels on the figure [?]. The construction is shown in Figure 1.

The same picture exposes the central geometry. Since $AC = CB$, the parallelograms AL and CH have equal content by I.36. The diagonal construction makes CH and HF complementary parallelograms, so I.43 gives their equality. Thus $AL = HF$. Adding the same rectangle CM identifies the whole rectangle AM with the gnomon NOP .

3 Classical Proof

The classical proof can be organized into four content steps.

First, the midpoint condition gives

$$AC = CB.$$

The parallelograms AL and CH stand on these equal bases and lie between the same parallels, so I.36 yields

$$AL = CH.$$

Second, CH and HF are complements about the diagonal DE of the relevant parallelogram configuration, hence I.43 gives

$$CH = HF.$$

Therefore

$$AL = HF.$$

4 The Geometric Identity in the Current Formalization

The public theorem receives

```
(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hABD : Geo.Between A B D)
```

so C is the midpoint of AB and B lies between A and D . The midpoint hypothesis gives

$$A - C - B \quad \text{and} \quad AC \cong CB.$$

Combining $A - C - B$ with $A - B - D$ by Hilbert betweenness transitivity yields

$$C - B - D \quad \text{and} \quad A - C - D.$$

Thus the full order

$$A - C - B - D$$

is derived, not read from a diagram.

At the exact-scissors level the public target is

$$\text{Sq}(BC) + \text{Rect}(AD, BD) \simeq_{\text{sc}} \text{Sq}(CD).$$

The order of the two summands on the left is the one present in the actual Lean conclusion. The theorem proves `HilbertScissorsEq`; it does not merely prove the weaker general content relation and it does not assert an equality of real numbers.

A second formal distinction is important. The square on BC in the public conclusion is the concrete square BCS_0S_1 . The rectangle contained by AD, BD is represented by the supplied quadrilateral $P_0P_1P_2P_3$. The statement is therefore about concrete representatives whose side magnitudes satisfy the required contained-by predicates.

5 Reusable Book II Infrastructure Used

5.1 Proposition II.3: extend the unequal-parts rectangle

The first major exact dissection is obtained by applying `euclid_proposition_2_3` to AD cut at B , with BD as the fixed second side:

$$\text{Rect}(AD, BD) \simeq_{\text{sc}} \text{Rect}(AB, BD) + \text{Sq}(BD).$$

This is the external-section analogue of introducing a square term in the earlier Book II identities.

5.2 Proposition II.1: split the old segment at its midpoint

The binary theorem `euclid_proposition_2_1_two` is then applied to AB cut at C , again with BD as the fixed side:

$$\text{Rect}(AB, BD) \simeq_{\text{sc}} \text{Rect}(AC, BD) + \text{Rect}(CB, BD).$$

Composing with the II.3 decomposition gives

$$\text{Rect}(AD, BD) \simeq_{\text{sc}} (\text{Rect}(AC, BD) + \text{Rect}(CB, BD)) + \text{Sq}(BD).$$

5.3 Midpoint transport

The midpoint congruence $AC \cong CB$ is converted into equality of rectangle content by `rectangle_transport_scissorsEq`:

$$\text{Rect}(AC, BD) \simeq_{\text{sc}} \text{Rect}(CB, BD).$$

Consequently

$$\text{Rect}(AD, BD) \simeq_{\text{sc}} (\text{Rect}(CB, BD) + \text{Rect}(CB, BD)) + \text{Sq}(BD).$$

No numerical factor 2 is introduced. The phrase “twice the rectangle” means two scissors terms with the same geometric content.

5.4 Proposition II.4 on the produced segment

The decisive second decomposition calls `euclid_proposition_2_4` directly on CD cut at B . The formal substitution is

$$A_{\text{II.4}} := C, \quad B_{\text{II.4}} := D, \quad C_{\text{II.4}} := B.$$

The required order hypothesis is exactly the derived betweenness $C - B - D$. The result is

$$\text{Sq}(CD) \simeq_{\text{sc}} (\text{Rect}(CB, BD) + \text{Sq}(BC)) + (\text{Rect}(CB, BD) + \text{Sq}(BD)).$$

This use of II.4 is one of the sharpest differences between II.5 and II.6 in the current formal library. II.5 imported the II.4 module but did not call `euclid_proposition_2_4`; II.6 calls it explicitly.

6 Proof Architecture of the Reconstruction

The production file preserves the names of the private incremental lemmas used during proof development. Their `testXX` names are implementation history only; because the declarations are `private`, they do not form part of the public API.

6.1 Step 1: recover the full order

`proposition2_6_test01_order` derives

$$C - B - D, \quad A - C - D, \quad AC \cong CB, \quad CA \cong CB.$$

The only input is the midpoint structure together with $A - B - D$.

6.2 Step 2: decompose the rectangle on AD, BD

`proposition2_6_test02_ad_decomposition` calls II.3 and proves

$$\text{Rect}(AD, BD) \simeq_{\text{sc}} \text{Rect}(AB, BD) + \text{Sq}(BD).$$

6.3 Step 3: split AB at C

`proposition2_6_test03_ab_decomposition` calls the binary II.1 theorem:

$$\text{Rect}(AB, BD) \simeq_{\text{sc}} \text{Rect}(AC, BD) + \text{Rect}(CB, BD).$$

6.4 Step 4: compose the two decompositions

`proposition2_6_test04_compose_decompositions` uses the same concrete representative of $\text{Rect}(AB, BD)$ in both preceding lemmas. Therefore no representative-uniqueness transport is needed at this junction. The result is

$$\text{Rect}(AD, BD) \simeq_{\text{sc}} (\text{Rect}(AC, BD) + \text{Rect}(CB, BD)) + \text{Sq}(BD).$$

6.5 Step 5: use the midpoint congruence

`proposition2_6_test05_midpoint_rectangle_transport` proves

$$\text{Rect}(AC, BD) \simeq_{\text{sc}} \text{Rect}(CB, BD)$$

by transporting the first adjacent side along $AC \cong CB$ while the second side remains congruent to BD .

6.6 Step 6: normalize the left decomposition

`proposition2_6_test06_double_cross` inserts the midpoint transport into Step 4:

$$\boxed{\text{Rect}(AD, BD) \simeq_{\text{sc}} (\text{Rect}(CB, BD) + \text{Rect}(CB, BD)) + \text{Sq}(BD).}$$

The proof uses only addition, reflexivity, and transitivity of exact scissors equality.

6.7 Step 7: expand the square on CD by II.4

`proposition2_6_test07_square_cd_decomposition` applies II.4 to the segment CD cut at B :

$$\boxed{\text{Sq}(CD) \simeq_{\text{sc}} (\text{Rect}(CB, BD) + \text{Sq}(BC)) + (\text{Rect}(CB, BD) + \text{Sq}(BD)).}$$

A useful implementation decision is visible here. The final assembly supplies the *same* rectangle representative $Z_0Z_1Z_2Z_3$ to both occurrences of $\text{Rect}(CB, BD)$, and it reuses the same supplied square on BD as in Step 6. Hence the following algebraic bridge can match the two decompositions literally.

6.8 Step 8: the abstract scissors bridge

`proposition2_6_test08_scissors_bridge` is purely algebraic at the level of formal scissors sums. For terms P, Q, R, S, T , it assumes

$$P \simeq (R + R) + T, \quad Q \simeq (R + S) + (R + T),$$

and proves

$$S + P \simeq Q.$$

The only rearrangement is associativity and commutativity of the formal sum, implemented by `ac_rfl`. The scissors equalities themselves are combined by `HilbertScissorsEq.add`, reflexivity, symmetry, and transitivity.

There is no cancellation theorem here. In particular, the proof does not infer equality by subtracting the common copies of R or T . It simply adds and rearranges exact scissors equalities.

6.9 Step 9: assemble II.6

`proposition2_6_test09_assembly` makes the literal substitution

$$\begin{aligned} P &:= \text{Rect}(AD, BD), \\ Q &:= \text{Sq}(CD), \\ R &:= \text{Rect}(CB, BD), \\ S &:= \text{Sq}(BC), \\ T &:= \text{Sq}(BD), \end{aligned}$$

into the abstract bridge. Steps 6 and 7 are exactly its two hypotheses, so the conclusion is the proposition:

$$\text{Sq}(BC) + \text{Rect}(AD, BD) \simeq_{\text{sc}} \text{Sq}(CD).$$

7 Lean Formalization

7.1 The public theorem signature

The following is the current compiling public signature, copied from the production source. It is intentionally explicit: the theorem receives the concrete cut diagrams and rectangle/square representatives required by the reused Book II APIs.

```
theorem euclid_proposition_2_6
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (A B C D : Geo.Point)
  (hMidC : HilbertIsMidpoint Geo C A B)
  (hABD : Geo.Between A B D)

  -----
  -- Data for test 06.
  -----

  (E2 H2 L2 X2 : Geo.Point)
  (hRect2 : IsRectangle Geo A D H2 E2)
  (hDH_BD2 : Geo.Congruent D H2 B D)
  (hELE2 : Geo.Between E2 L2 H2)
  (hEXD2 : Geo.Between E2 X2 D)
  (hBXL2 : Geo.Between B X2 L2)
  (hLeftPar2 : IsParallelogram Geo L2 E2 A B)
  (hRightPar2 : IsParallelogram Geo D H2 L2 B)
```

```

-- Square on BD, shared by tests 06 and 07.
(F G : Geo.Point)
(hSquareBD : IsSquare Geo B D F G)

-- Rect(AD,BD).
(P0 P1 P2 P3 : Geo.Point)
(hAD_BD :
  IsRectangleContainedBy Geo
    P0 P1 P2 P3 A D B D)

-- Rect(AB,BD).
(V0 V1 V2 V3 : Geo.Point)
(hAB_BD :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 A B B D)

(E3 H3 L3 X3 : Geo.Point)
(hRect3 : IsRectangle Geo A B H3 E3)
(hBH_BD3 : Geo.Congruent B H3 B D)
(hELE3 : Geo.Between E3 L3 H3)
(hEXB3 : Geo.Between E3 X3 B)
(hCXL3 : Geo.Between C X3 L3)
(hLeftPar3 : IsParallelogram Geo L3 E3 A C)
(hRightPar3 : IsParallelogram Geo B H3 L3 C)

-- Rect(AC,BD).
(Y0 Y1 Y2 Y3 : Geo.Point)
(hAC_BD :
  IsRectangleContainedBy Geo
    Y0 Y1 Y2 Y3 A C B D)

-- The single representative R = Rect(CB,BD).
(Z0 Z1 Z2 Z3 : Geo.Point)
(hCB_BD :
  IsRectangleContainedBy Geo
    Z0 Z1 Z2 Z3 C B B D)

-----

-- Data for test 07: II.4 on CD cut at B.
-----

(D0 E0 L0 X0 : Geo.Point)
(hSquareCD : IsSquare Geo C D E0 D0)

(hD0LOEO : Geo.Between D0 L0 E0)
(hD0X0D : Geo.Between D0 X0 D)
(hBX0LO : Geo.Between B X0 L0)

(hII4LeftPar :
  IsParallelogram Geo L0 D0 C B)

(hII4RightPar :
  IsParallelogram Geo D E0 L0 B)

-- Representatives required internally by II.4.
(J0 J1 J2 J3 : Geo.Point)
(K0 K1 K2 K3 : Geo.Point)

```



```

(hCD_CB :
  IsRectangleContainedBy Geo
    J0 J1 J2 J3 C D C B)

(hCD_BD :
  IsRectangleContainedBy Geo
    K0 K1 K2 K3 C D B D)

-----
-- Left II.3 decomposition inside II.4.
-----

(D1 E1 L1 X1 : Geo.Point)

(hRectLeft :
  IsRectangle Geo D C E1 D1)

(hCE1_BC :
  Geo.Congruent C E1 B C)

(hD1L1E1 : Geo.Between D1 L1 E1)
(hD1X1C : Geo.Between D1 X1 C)
(hBX1L1 : Geo.Between B X1 L1)

(hLeftLeftPar :
  IsParallelogram Geo L1 D1 D B)

(hLeftRightPar :
  IsParallelogram Geo C E1 L1 B)

-- Square on BC.
(S0 S1 : Geo.Point)
(hSquareBC : IsSquare Geo B C S0 S1)

-----
-- Right II.3 decomposition inside II.4.
-----

(D2 E4 L4 X4 : Geo.Point)

(hRectRight :
  IsRectangle Geo C D E4 D2)

(hDE4_BD :
  Geo.Congruent D E4 B D)

(hD2L4E4 : Geo.Between D2 L4 E4)
(hD2X4D : Geo.Between D2 X4 D)
(hBX4L4 : Geo.Between B X4 L4)

(hRightLeftPar :
  IsParallelogram Geo L4 D2 C B)

(hRightRightPar :
  IsParallelogram Geo D E4 L4 B) :

HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo B C S0 S1 +

```

```

    hilbertParallelogramTerm Geo P0 P1 P2 P3)
(hilbertParallelogramTerm Geo C D E0 D0)

```

The large signature falls into three geometric groups.

1. The points A, B, C, D , midpoint data, and $A - B - D$ encode the baseline configuration.
2. The first block supplies the two exact decompositions used to reach Step 6: II.3 on AD cut at B , II.1 on AB cut at C , a square on BD , and representatives of the three required rectangles.
3. The second block supplies the II.4 diagram on CD cut at B , including its two internal II.3 configurations, the square on BC , and the intermediate rectangle representatives required by II.4.

The conclusion itself is much smaller than the witness data:

```

HilbertScissorsEq Geo
(hilbertParallelogramTerm Geo B C S0 S1 +
  hilbertParallelogramTerm Geo P0 P1 P2 P3)
(hilbertParallelogramTerm Geo C D E0 D0)

```

and is exactly the formal version of

$$\text{Sq}(BC) + \text{Rect}(AD, BD) = \text{Sq}(CD).$$

7.2 The final bridge call

After the two large decompositions have been established, the conceptual end of the proof is only the following substitution into the abstract bridge:

```

exact
  proposition2_6_test08_scissors_bridge
    Geo
      (hilbertParallelogramTerm Geo P0 P1 P2 P3)
      (hilbertParallelogramTerm Geo C D E0 D0)
      (hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)
      (hilbertParallelogramTerm Geo B C S0 S1)
      (hilbertParallelogramTerm Geo B D F G)
    h06
    h07

```

This makes the proof architecture unusually transparent: geometry produces two normal forms, and the last step is an exact-scissors rearrangement of those normal forms.

8 Relationship with Earlier Book II Propositions

8.1 Proposition II.1

II.1 is called directly through `euclid_proposition_2_1_two`. Its role is to split $\text{Rect}(AB, BD)$ at the midpoint C . Thus the basic rectangle-distribution theorem remains the additive engine of Book II.

8.2 Proposition II.3

II.3 is also called directly. Applied to AD cut at B , it introduces the square on the added segment BD :

$$\text{Rect}(AD, BD) = \text{Rect}(AB, BD) + \text{Sq}(BD)$$

at the exact-scissors level.

8.3 Proposition II.4

II.4 is a direct theorem dependency and supplies the expansion of the square on CD after B has been proved to lie between C and D . Since the current II.4 theorem itself composes earlier Book II infrastructure, II.6 inherits that established internal structure rather than rebuilding it.

8.4 Proposition II.5

The production file imports `CGJteamLab.Proposition2_5`, so the II.5 module lies immediately below II.6 in the module graph. Nevertheless `euclid_proposition_2_5` is not called by any of the II.6 private helpers or by the public theorem.

Mathematically the two propositions are companions: II.5 concerns an unequal cut inside a bisected segment, whereas II.6 concerns an extension beyond the endpoint. Formally their proof DAGs differ. The current II.5 proof does not call II.4, while II.6 uses II.4 directly as one of its two main normal-form decompositions.

9 Relationship with Book I Infrastructure

The classical proof uses I.36 and I.43 explicitly. The Lean proof does not replay those steps proposition-locally. Their geometric effect has already been absorbed into the rectangle, parallelogram, square, and scissors infrastructure on which II.1–II.4 are built.

Book I geometry still remains foundational below the current theorem: right-angle propagation, squares, parallelograms, parallel lines, and the finite-scissors representation all depend on the established Hilbert/Book I layers. The gain is architectural: II.6 can be expressed as a composition of stable Book II content interfaces rather than as a second formalization of Euclid’s gnomon diagram.

10 Formalization Notes and Hidden Geometric Data

10.1 The extension order is proved

The diagram suggests $A - C - B - D$, but the public theorem assumes only midpoint data for C together with $A - B - D$. The private order lemma explicitly proves $C - B - D$ and $A - C - D$. This matters because II.4 can be applied to CD cut at B only after $C - B - D$ is available as a theorem.

10.2 Midpoint data have two independent roles

`HilbertIsMidpoint` is not a numerical assertion that C has coordinate one half. Its betweenness component supplies $A - C - B$, while its congruence component supplies $AC \cong CB$. The former feeds the order DAG; the latter feeds rectangle transport.

10.3 Representative reuse is deliberate

In the final II.4 call the same Z representative is used twice for $\text{Rect}(CB, BD)$, and the same square $BDFG$ is shared with the Step 6 decomposition. This is not a mathematical restriction on the content identity. It is a deliberate proof-engineering choice that lets the final abstract scissors bridge identify the repeated terms literally and avoids unnecessary representative transports.

10.4 The square terms remain geometric representatives

The terms called $\text{Sq}(BC)$, $\text{Sq}(BD)$, and $\text{Sq}(CD)$ are concrete squares supplied through `IsSquare`. They are inserted into the scissors calculus as parallelogram terms. No operation sends a segment to a real number or squares a numerical length.

10.5 No gnomon primitive

As in II.5, the formal library does not introduce a special gnomon type. Euclid's shaded remainder is represented extensionally by a formal sum of rectangle and square terms. In II.6 the combination of II.4 with the abstract scissors bridge replaces the gnomon bookkeeping completely.

10.6 No cancellation or subtraction of content

The identity strongly resembles a difference-of-squares calculation, but the proof does not subtract a square or cancel equal rectangles. The abstract bridge is additive: it uses only exact scissors equality, addition, associativity/commutativity of the formal sum, symmetry, and transitivity.

11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`. Both the module build and the full project build were reported clean.

The user-verified transitive axiom audit is

```
#print axioms Geometry.euclid_proposition_2_6

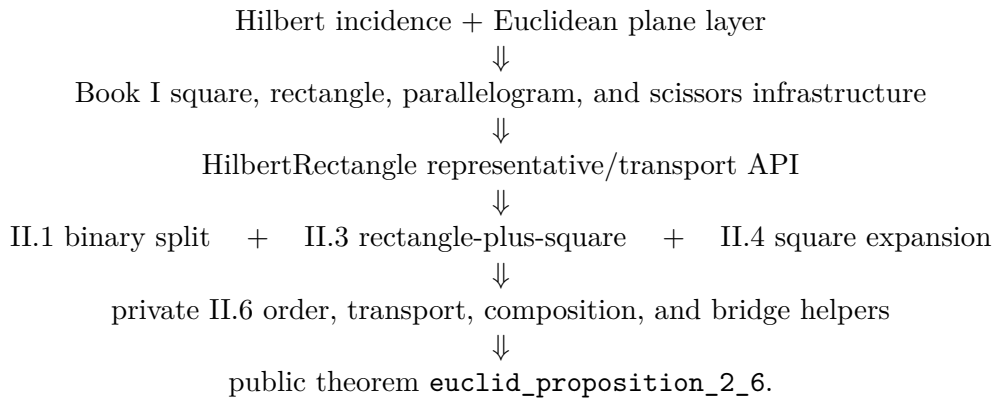
'Geometry.euclid_proposition_2_6' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

The precise status is therefore not “axiom-free.” Rather, II.6 introduces no new proposition-specific axiom and no `sorry`; its reported transitive dependency list is exactly the four declarations above and contains no `sorryAx`.

Semantically, the theorem lies in the exact finite-scissors layer. It proves `HilbertScissorsEq` for the supplied configurations. It does not invoke a numerical area function, a segment multiplication operation, or a general cancellation principle for equal content.

12 Dependency Summary

The actual theorem-call DAG can be displayed as



The module chain is slightly different:

Proposition2_5 module $\longrightarrow$ Proposition2_6 module,

but `euclid_proposition_2_5` itself is not called.

The classical local chain is different again:

I.36 + I.43 $\longrightarrow$ gnomon identification $\longrightarrow$ add square on BC $\longrightarrow$ II.6.

The formal reconstruction therefore proves the same geometric-content identity through a different DAG rather than translating Euclid line by line.

New geometric axiom in II.6:	no
New content axiom in II.6:	no
New proposition-local helper theorem:	yes, nine private helpers
New reusable public helper theorem:	no
New <code>HilbertRectangle</code> theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.1, II.3, II.4
Imports II.5 module:	yes
Calls <code>euclid_proposition_2_5</code> :	no
Uses exact <code>HilbertScissorsEq</code> :	yes
Uses general equicomplementability:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full project build:	clean (user verified)
Proposition-local <code>sorry/admit</code> :	no
Transitive <code>sorryAx</code> :	absent in reported audit

13 Conclusion

Proposition II.6 is a strong checkpoint for the reuse of the Book II rectangle calculus. Classically, Euclid proves the identity by constructing a gnomon inside the square on CD and comparing its pieces through I.36 and I.43. The current Lean proof reaches the same content equality without a gnomon primitive and without numerical algebra.

Its architecture is instead a composition of three already established Book II mechanisms: II.3 decomposes the rectangle on AD, BD , II.1 splits the old segment AB at its midpoint, and II.4 expands the square on CD at the point B . Midpoint congruence normalizes the repeated cross rectangles, and a small abstract scissors lemma performs the final associative-commutative rearrangement. The result is an exact `HilbertScissorsEq` proof with no new axiom, no `sorry`, no segment multiplication, and no numerical area.